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Étale cohomology of curves

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Introduction

This post grew out of a theorem I encountered last semester in a course on étale cohomology. I wanted to collect in one place the ingredients entering the proof, mostly for my own understanding.

The theorem we prove is the following.

Theorem 0.1. *Let C be an algebraic curve (of dimension 1, finite type, and separated over a field) over a separably closed field k . Then*

$$H^i(C, \mathcal{O}_{C, \text{ét}}^\times) = \begin{cases} \mathcal{O}_C(C)^\times & i = 0 \\ \text{Pic}(C) & i = 1 \\ 0 & i > 1 \text{ and } k \text{ is algebraically closed} \\ p^\infty\text{-torsion group} & i > 1 \text{ and } k \text{ is non-perfect with } \text{char}(k)=p \end{cases}$$

One of the crucial points of the proof is to reduce the computation of higher cohomology of $\mathcal{O}_{C, \text{ét}}^\times$ to the generic point of the curve. Once this is done, the problem becomes a statement in Galois cohomology.

Preliminaries

In this section we collect the auxiliary facts used in the proof: vanishing results for finite schemes over a field, finite sums of skyscraper sheaves, and torsion sheaves.

Lemma 0.1. *Let X be a finite scheme over a separably closed field k . Then $H^i(X, \mathcal{F}) = 0$ for $i > 0$ and for all $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$.*

Proof. Since the structure map $f : X \rightarrow \operatorname{Spec}(k)$ is finite, $f_* : \operatorname{Sh}(X_{\text{ét}}) \rightarrow \operatorname{Sh}(\operatorname{Spec}(k)_{\text{ét}})$ is exact. Therefore, the derived functor $R^q f_*$ vanishes for every $q > 0$.

Consider the Grothendieck spectral sequence

$$E_2^{p,q} = H^p(\operatorname{Spec}(k)_{\text{ét}}, R^q f_* \mathcal{F}) \implies H^{p+q}(X_{\text{ét}}, \mathcal{F})$$

By the vanishing of $R^q f_*$ we get that $H^p(\operatorname{Spec}(k)_{\text{ét}}, f_* \mathcal{F}) \cong H^p(X_{\text{ét}}, \mathcal{F})$. Since k is separably closed, it is a strictly henselian ring, so global sections are isomorphic to stalks. Hence global sections are exact, and the cohomology vanishes. $\square$

Lemma 0.2. *Let X be a scheme, $i : Z \rightarrow X$ a closed immersion, and $\mathcal{F} \in \operatorname{Sh}(X_{\text{ét}})$ be supported on Z . Then the canonical morphism $\eta : \mathcal{F} \rightarrow i_* i^* \mathcal{F}$ is an isomorphism.*

Proof. We check that η is an isomorphism on stalks. Let $\bar{x}$ be a geometric point of X , and let x be the underlying point. Since i is a closed immersion, it is finite. Hence, by Stacks project [Tag 0DK3](#), we have

$$(i_* i^* \mathcal{F})_{\bar{x}} \cong \prod_{\bar{z}} (i^* \mathcal{F})_{\bar{z}},$$

where the product is taken over the geometric points $\bar{z}$ of Z lying over $\bar{x}$.

If $x \notin Z$, this product is empty, hence

$$(i_* i^* \mathcal{F})_{\bar{x}} = 0.$$

Since $\mathcal{F}$ is supported on Z , also $\mathcal{F}_{\bar{x}} = 0$, so $\eta_{\bar{x}}$ is an isomorphism.

If $x \in Z$, then $\bar{x}$ factors uniquely through Z , say as a geometric point $\bar{z}$ of Z . Hence the product has exactly one factor, and

$$(i_* i^* \mathcal{F})_{\bar{x}} \cong (i^* \mathcal{F})_{\bar{z}} \cong \mathcal{F}_{\bar{x}}$$

$\square$

Corollary 0.1 (of Lemma 0.2). *Let X be a scheme and $\mathcal{F} \in \operatorname{Sh}(X_{\text{ét}})$ supported on finitely many closed points of X . Then*

$$\mathcal{F} \cong \bigoplus_{x \in \operatorname{supp}(\mathcal{F})} x_* \mathcal{F}_x$$

Proof. We apply Lemma 0.2 with $Z := \{x \rightarrow X : x \text{ is a geometric point}\}$ and $i : Z \rightarrow X$. We then have that

$$\mathcal{F} \cong i_* i^* \mathcal{F}$$

and

$$i_* i^* \mathcal{F} \cong i_* \left(\bigoplus_{x \in \operatorname{supp}(X)} \mathcal{F}_x \right) \cong \bigoplus_{x \in \operatorname{supp}(X)} x_* \mathcal{F}_x$$

where the latter isomorphism holds because the points are finitely many. $\square$

Lemma 0.3. *Let X be a scheme, x be a geometric point of X , and $\mathcal{F} \in \text{Sh}(x_{\text{ét}})$. Then for every $i > 0$*

$$R^i x_* \mathcal{F} = 0.$$

In particular,

$$H^i(X_{\text{ét}}, x_* \mathcal{F}) = 0$$

for all $i > 0$.

Proof. Consider an exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$$

of étale sheaves over $x_{\text{ét}}$. Since x is a geometric point, the functor x_* is exact. Indeed, sections of $x_* \mathcal{F}$ over an étale $U \rightarrow X$ are products of stalks over the geometric fibre of U at x , and products of abelian groups are exact. Therefore

$$R^i x_* \mathcal{F} = 0$$

for every $i > 0$.

As a consequence, with the same reasoning used in the proof of Lemma 0.1 using the Grothendieck spectral sequence, we obtain

$$H^i(X_{\text{ét}}, x_* \mathcal{F}) = 0$$

for every $i > 0$. □

Lemma 0.4. *Let X be a quasi-compact quasi-separated (qcqs) scheme, and $\mathcal{F} \in \text{Sh}(X_{\text{ét}})$ be a torsion sheaf. Then $H^i(X, \mathcal{F})$ is a torsion group for all $i \geq 0$.*

Proof. Let $\mathcal{F}[n]$ be the kernel of the multiplication by n map $\mathcal{F} \xrightarrow{\cdot n} \mathcal{F}$. We have an isomorphism

$$\varinjlim (\mathcal{F}[n]) \rightarrow \mathcal{F}$$

because it is an isomorphism on stalks:

$$(\varinjlim \mathcal{F}[n])_x \cong \varinjlim ((\mathcal{F}[n])_x) \cong \varinjlim ((\mathcal{F}_x)[n]) \cong \mathcal{F}_x$$

Since X is qcqs, the cohomology commutes with colimits:

$$H^i(X_{\text{ét}}, \mathcal{F}) \cong H^i(X_{\text{ét}}, \varinjlim (\mathcal{F}[n])) \cong \varinjlim (H^i(X_{\text{ét}}, \mathcal{F}[n]))$$

Since the multiplication by n map is the zero map on $\mathcal{F}[n]$, in cohomology it is again the zero map (since the cohomology functor is additive). Therefore, the cohomology is a colimit of torsion groups, and therefore it is a torsion group. □

Lemma 0.5. *Let k be a separably closed field, K/k a field extension of transcendence degree 1, and L/K be an algebraic and separable field extension. Then $L' := (L \otimes_k \bar{k})_{\text{red}}$, where $\bar{k}$ denotes an algebraic closure, is a field with C_1 property and L'/L is purely inseparable.*

Proof. We first show that L'/L is purely inseparable. If $\text{char}(k) = 0$, then k is already algebraically closed, so $\bar{k} = k$, hence $L' = L$, and there is nothing to prove. If $\text{char}(k) = p > 0$, it is easy to verify that for every $x \in L \otimes_k \bar{k}$ we get for a suitable n that $x^{p^n} \in L$ (write x as a sum of elementary tensor). Thus, passing to the reduction gives the purely inseparable extension.

It remains to prove that L' is C_1 . Since $L/k(t)$ is algebraic, $A := L \otimes_k \bar{k}$ is integral over $k(t) \otimes_k \bar{k}$. Now t is transcendental over k , so

$$k(t) \otimes_k \bar{k} \cong \bar{k}(t).$$

Hence A is integral over the field $\bar{k}(t)$, and therefore $L'/\bar{k}(t)$ is algebraic (being L' a quotient of A).

Now $\bar{k}$ is algebraically closed, so by [Tsen's theorem](#) the field $\bar{k}(t)$ is C_1 . Since algebraic extensions of C_1 -fields are again C_1 , it follows that L' is C_1 . $\square$

Proof of the theorem

We are now ready to prove that

$$H^i(C, \mathcal{O}_{C, \text{ét}}^\times) = \begin{cases} \mathcal{O}_C(C)^\times & i = 0 \\ \text{Pic}(C) & i = 1 \\ 0 & i > 1 \text{ and } k \text{ is algebraically closed} \\ p^\infty\text{-torsion group} & i > 1 \text{ and } k \text{ is non-perfect with } \text{char}(k)=p \end{cases}$$

Proof. The case $i = 0$ is clear. We will not prove the case $i = 1$, since it follows from the interpretation of first cohomology in terms of torsors. More precisely, $H^1(C_{\text{ét}}, \mathcal{O}_{C, \text{ét}}^\times)$ classifies $\mathcal{O}_{C, \text{ét}}^\times$ -torsors on $C_{\text{ét}}$, see [Tag 03AJ](#). These torsors are equivalent to line bundles on C , thus we obtain

$$H^1(C_{\text{ét}}, \mathcal{O}_{C, \text{ét}}^\times) \cong \text{Pic}(C),$$

see [Tag 040E](#) for the proof.

We now study the case of $i > 1$.

- **Step 1:** suppose the curve is regular and connected.

In particular, the curve is integral, since every stalk is a noetherian regular local ring (hence reduced) and since every point belongs to exactly one irreducible component. Since irreducible components are closed (because C is a noetherian topological space), the curve is indeed integral.

Let η be the generic point. Consider the following exact sequence:

$$0 \rightarrow \mathcal{O}_{C,\text{ét}}^\times \rightarrow \eta_* \mathcal{O}_{\eta,\text{ét}}^\times \rightarrow \bigoplus_{c \in C \text{ closed}} c_* \mathbb{Z} \rightarrow 0 \quad (1)$$

where the second map is the divisor map. This sequence is exact since on the stalk at a geometric point $\bar{x}$ over a closed point it is given by

$$0 \rightarrow (\mathcal{O}_{C,\bar{x}}^{\text{sh}})^\times \rightarrow (\text{Frac}(\mathcal{O}_{C,\bar{x}}^{\text{sh}}))^\times \rightarrow \mathbb{Z} \rightarrow 0$$

and the right-hand side map is the valuation map.

Taking cohomology of sequence 1 gives that

$$H^i(C, \mathcal{O}_{C,\text{ét}}^\times) \cong H^i(C, \eta_* \mathcal{O}_{\eta,\text{ét}}^\times) \quad (2)$$

for all $i > 1$. Indeed, the sheaf $\bigoplus_{c \in C \text{ closed}} c_* \mathbb{Z}$ is the filtered colimit of the finite sums $\bigoplus_{c \in S} c_* \mathbb{Z}$, where S ranges over the finite subsets of closed points of C . By Corollary 0.1 and Lemma 0.3, each such finite sum has vanishing higher cohomology. Since C is qcqs, filtered colimits of abelian sheaves on $C_{\text{ét}}$ commute with cohomology by the Stacks Project Tag 073D. Therefore

$$H^i \left(C_{\text{ét}}, \bigoplus_{c \in C \text{ closed}} c_* \mathbb{Z} \right) = 0$$

for every $i > 0$, and the claimed isomorphism follows from the long exact sequence in cohomology.

By Tag 03Q9, we have for every geometric point c of C

$$(R^q \eta_* \mathcal{O}_{\eta,\text{ét}}^\times)_c \cong H^q((\eta \times_C \text{Spec}(\mathcal{O}_{C,c}^{\text{sh}}))_t, (\mathcal{O}_{\eta,\text{ét}}^\times)_{|\eta \times_C \text{Spec}(\mathcal{O}_{C,c}^{\text{sh}})}) \quad (3)$$

As $\eta \times_C \text{Spec}(\mathcal{O}_{C,c}^{\text{sh}}) = \text{Spec}(\text{Frac}(\mathcal{O}_{C,c}^{\text{sh}}))$, write L for $\text{Frac}(\mathcal{O}_{C,c}^{\text{sh}})$, so that the restriction of $\mathcal{O}_{\eta,\text{ét}}^\times$ appearing in the right-hand side of 3 is just $\mathcal{O}_{\text{Spec}(L),\text{ét}}^\times$. Observe that L is an algebraic separable extension of the function field $\kappa(C)$. Indeed, $\mathcal{O}_{C,c}^{\text{sh}}$ is the filtered colimit of local étale $\mathcal{O}_{C,c}$ -algebras B , and for each such B the field $\text{Frac}(B) = B \otimes_{\mathcal{O}_{C,c}} \kappa(C)$ is a finite separable extension of $\kappa(C)$, thus every element of L lies in some finite separable extension of $\kappa(C)$.

By the [relation between Galois and étale cohomology](#), we have

$$H^q(\mathrm{Spec}(L), \mathcal{O}_{\mathrm{Spec}(L), \mathrm{ét}}^\times) \cong H^q(\mathrm{Gal}(L^{\mathrm{sep}}/L), (L^{\mathrm{sep}})^\times) \quad (4)$$

By Hilbert 90, this vanishes for $q = 1$.

Claim. The group

$$H^q(\mathrm{Gal}(L^{\mathrm{sep}}/L), (L^{\mathrm{sep}})^\times)$$

is a p^∞ -torsion group for every $q \geq 2$.

Proof. By Lemma 0.5, $L' := (L \otimes_k \bar{k})_{\mathrm{red}}$ is a C_1 -field with L'/L purely inseparable. It is a well-known Galois-cohomology fact (Stacks project [Tag 03R8](#) and [Tag 03RC](#)) that for a C_1 -field

$$H^i(\mathrm{Gal}(L'^{\mathrm{sep}}/L'), (L'^{\mathrm{sep}})^\times) = 0$$

for every $i > 0$. Therefore, by pure inseparability of L'/L , we get, by the isomorphism on Galois groups,

$$H^i(\mathrm{Gal}(L^{\mathrm{sep}}/L), (L^{\mathrm{sep}})^\times) \cong H^i(\mathrm{Gal}(L'^{\mathrm{sep}}/L'), (L'^{\mathrm{sep}})^\times) = 0$$

for every $i > 0$.

Consider the short exact sequence

$$0 \rightarrow (L^{\mathrm{sep}})^\times \rightarrow (L'^{\mathrm{sep}})^\times \rightarrow Q \rightarrow 0$$

where Q is the cokernel of the first map. By pure inseparability of $L'^{\mathrm{sep}}/(L^{\mathrm{sep}})$, we have that Q is a p^∞ -torsion group. Passing to cohomology, we get

$$H^{i+1}(\mathrm{Gal}(L^{\mathrm{sep}}/L), (L^{\mathrm{sep}})^\times) \cong H^i(\mathrm{Gal}(L^{\mathrm{sep}}/L), Q)$$

for every $i > 0$. Therefore, by Lemma 0.4, we get the claim. $\square$

By isomorphisms 3, 4, and [Claim](#), we get that

$$(R^q \eta_* \mathcal{O}_{\eta, \mathrm{ét}}^\times)_c$$

is a p^∞ -torsion group at every geometric point c , thus the sheaf $R^q \eta_* \mathcal{O}_{\eta, \mathrm{ét}}^\times$ is a p^∞ -torsion sheaf.

Consider the Grothendieck spectral sequence

$$E_2^{i,j} := H^i(C_{\mathrm{ét}}, R^j \eta_* \mathcal{O}_{\eta, \mathrm{ét}}^\times) \Rightarrow H^{i+j}(\eta_{\mathrm{ét}}, \mathcal{O}_{\eta, \mathrm{ét}}^\times).$$

Since this is a first-quadrant spectral sequence, we also get

$$E_2^{i,j} \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] \Rightarrow H^{i+j}(\eta_{\mathrm{ét}}, \mathcal{O}_{\eta, \mathrm{ét}}^\times) \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}].$$

For every $j \geq 1$, we proved that $R^j \eta_* \mathcal{O}_{\eta, \text{ét}}^\times$ is a p^∞ -torsion sheaf. Hence $E_2^{i,j} = H^i(C_{\text{ét}}, R^j \eta_* \mathcal{O}_{\eta, \text{ét}}^\times)$ is a p^∞ -torsion group for every $i \geq 0$ and every $j \geq 1$, by Lemma 0.4. Therefore

$$E_2^{i,j} \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] = 0 \quad (j \geq 1).$$

Thus the localized spectral sequence has only the row $j = 0$ possibly nonzero, and therefore it degenerates at the E_2 -page. Hence for every $n > 0$ one gets an isomorphism

$$E_2^{n,0} \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] \cong H^n(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times) \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}].$$

Now $H^1(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times) = 0$ by Hilbert-90, and for every $n \geq 2$ the group $H^n(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times)$ is a p^∞ -torsion group by the same Galois-cohomological argument as above, applied to the field $\kappa(C)$. Therefore

$$H^n(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times) \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] = 0 \quad (n > 0).$$

It follows that

$$E_2^{n,0} \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] = 0 \quad (n > 1).$$

Since $E_2^{n,0} = H^n(C_{\text{ét}}, \eta_* \mathcal{O}_{\eta, \text{ét}}^\times)$, we obtain

$$H^n(C_{\text{ét}}, \eta_* \mathcal{O}_{\eta, \text{ét}}^\times) \otimes_{\mathbb{Z}} \mathbb{Z}[p^{-1}] = 0 \quad (n > 1).$$

By the isomorphism 2, this gives the claim in the case of a regular connected curve.

If k is algebraically closed, the previous argument gives a stronger conclusion. Indeed, in this case the field $L = \text{Frac}(\mathcal{O}_{C,c}^{\text{sh}})$ is a C_1 -field, being an algebraic extension of the function field $\kappa(C)$. Hence, by the same Galois-cohomological vanishing used above, we have

$$H^q(\text{Gal}(L^{\text{sep}}/L), (L^{\text{sep}})^\times) = 0$$

for every $q > 0$. Therefore

$$(R^q \eta_* \mathcal{O}_{\eta, \text{ét}}^\times)_c = 0$$

for every geometric point c of C and every $q > 0$, so

$$R^q \eta_* \mathcal{O}_{\eta, \text{ét}}^\times = 0$$

for every $q > 0$.

Thus the Grothendieck spectral sequence degenerates and gives

$$H^n(C_{\text{ét}}, \eta_* \mathcal{O}_{\eta, \text{ét}}^\times) \cong H^n(\eta_{\text{ét}}, \mathcal{O}_{\eta, \text{ét}}^\times)$$

for every $n \geq 0$. Since $\kappa(C)$ is again a C_1 -field, the right-hand side vanishes for every $n > 0$. Therefore

$$H^n(C_{\text{ét}}, \eta_* \mathcal{O}_{\eta, \text{ét}}^\times) = 0$$

for every $n > 0$. By 2, this proves the vanishing of

$$H^n(C_{\text{ét}}, \mathcal{O}_{C, \text{ét}}^\times)$$

for every $n > 1$ in the case where k is algebraically closed.

- **Step 2:** suppose C is regular.

Apply **Step 1** to every connected component.

- **Step 3:** suppose C is reduced.

Consider the normalization $\nu : C' \rightarrow C$ of the curve. We first prove that the natural morphism

$$\mathcal{O}_{C, \text{ét}}^\times \rightarrow \nu_* \mathcal{O}_{C', \text{ét}}^\times$$

is injective.

Let C_{reg} be the regular locus of C . It is Zariski open in C , since C is locally of finite type over a field. Moreover, since C is reduced, each generic point of C is regular, thus C_{reg} is dense. Since regular points are normal, the normalization is an isomorphism over C_{reg} . Hence

$$(\mathcal{O}_{C, \text{ét}}^\times)_{|C_{\text{reg}}} \rightarrow (\nu_* \mathcal{O}_{C', \text{ét}}^\times)_{|C_{\text{reg}}}$$

is an isomorphism.

Now let $V \rightarrow C$ be an affine étale morphism. Since étale morphisms are open,

$$V_{\text{reg}} := V \times_C C_{\text{reg}}$$

is a dense open subset of V . After base change to V , the normalization map is still an isomorphism over V_{reg} . Therefore the map

$$\Gamma(V, \mathcal{O}_V^\times) \rightarrow \Gamma(V \times_C C', \mathcal{O}_{V \times_C C'}^\times)$$

becomes an isomorphism after restricting to the dense open subset V_{reg} .

Since C is reduced and $V \rightarrow C$ is étale, also V is reduced. Hence restriction to a dense open subset is injective, thus

$$\Gamma(V, \mathcal{O}_V^\times) \rightarrow \Gamma(V \times_C C', \mathcal{O}_{V \times_C C'}^\times)$$

is injective. Since this holds for every affine étale $V \rightarrow C$, the morphism

$$\mathcal{O}_{C, \text{ét}}^\times \rightarrow \nu_* \mathcal{O}_{C', \text{ét}}^\times$$

is injective.

Now define Q as the cokernel of this injective morphism. We have an exact sequence

$$0 \rightarrow \mathcal{O}_{C, \text{ét}}^\times \rightarrow \nu_* \mathcal{O}_{C', \text{ét}}^\times \rightarrow Q \rightarrow 0 \quad (5)$$

Since the normalization is an isomorphism over C_{reg} , the sheaf Q vanishes on C_{reg} . Hence Q is supported on finitely many points of C . Therefore, by Corollary 0.1, we get

$$H^i(C_{\text{ét}}, \mathcal{O}_{C, \text{ét}}^\times) \cong H^i(C_{\text{ét}}, \nu_* \mathcal{O}_{C', \text{ét}}^\times)$$

for every $i \geq 2$.

Moreover, by the Stacks Project Tag 035S and Tag 035B, ν is finite. Hence, by the exactness of the direct image of a finite morphism on étale sheaves, ν_* is exact, giving

$$H^i(C_{\text{ét}}, \nu_* \mathcal{O}_{C', \text{ét}}^\times) \cong H^i(C', \mathcal{O}_{C', \text{ét}}^\times)$$

Since C' is normal of dimension 1, it is regular, so we can apply **Step 2**.

- **Step 4:** no additional hypotheses on C .

Consider $\mathcal{N} := \text{Nil}(\mathcal{O}_C)$. We proceed by induction on n such that $\mathcal{N}^{2^n} = 0$. The case $n = 0$ is given by **Step 3**. Assume now that $n > 0$ and that the result is known whenever the nilradical is killed by 2^{n-1} . Define $J := \mathcal{N}^{2^{n-1}}$, let C_0 be the closed subscheme defined by it, and let $\pi : C_0 \rightarrow C$. Since $J^2 = \mathcal{N}^{2^n} = 0$, we get an exact sequence

$$0 \rightarrow J \rightarrow \mathcal{O}_{C, \text{ét}}^\times \rightarrow \pi_* \mathcal{O}_{C_0, \text{ét}}^\times \rightarrow 1$$

where the left map sends an element a to $1 + a$ on sections. Since J is quasi-coherent,

$$H^i(C_{\text{ét}}, J) \cong H^i(C_{\text{Zar}}, J)$$

for every $i \geq 0$. Thus, by dimension vanishing, we get an isomorphism

$$H^i(C_{\text{ét}}, \mathcal{O}_{C, \text{ét}}^\times) \cong H^i(C_{\text{ét}}, \pi_* \mathcal{O}_{C_0, \text{ét}}^\times)$$

for every $i > 1$.

Since π is a closed immersion, it is finite, thus π_* is exact, giving

$$H^i(C_{\text{ét}}, \pi_* \mathcal{O}_{C_0, \text{ét}}^\times) \cong H^i((C_0)_{\text{ét}}, \mathcal{O}_{C_0, \text{ét}}^\times)$$

for every $i \geq 0$. Moreover, the nilradical of $\mathcal{O}_{C_0}$ is killed by 2^{n-1} . Applying the induction hypothesis, we get the claim. □